

Supplementary Material:
ConstructionSceneReady: Evidence-grounded
compilation and task-scene validation for BIM-driven
robotic assembly

Abstract

This supplementary file collects the larger mapping, registry, design, ablation, and per-condition statistics tables, plus the worked case-study figure, referenced from the main manuscript. Table and figure numbers carry an “S” prefix and are cited from the main text as Table S1...S8 and Figure S9.

Table S1: Task-information mapping for the bolt-insertion assembly task. For each building-information family: the robot function it serves, the task phase in which it acts, and the measured consequence of its absence or error (Section 7.3 of the main text). “Scaffolded” marks families whose dynamical channel is removed by the virtual-grasp executor; their task consequence is explicitly *not* claimed by this study.

Information family	Robot served	function	Task phase	Measured consequence of defect
Semantic identity & task reference	Select which member, interface and hole to act on		target selection	<i>wrong</i> value discretely fatal (0%); <i>missing</i> label degrades (12.0% reward, 1.0% geo.); no perceptual compensation for a wrong target
Member pose (absolute)	Coarse approach and reach	approach planning	transport	largely compensated: reward -4.5 pp (significant on seed 42, $p=0.012$), geo. -2.0 pp (n.s.)
Interface coordinates relative to member	Fine alignment at the hole mouth		align, insert	reward-tolerance partly masks it (-1.0 pp at $2-5$ mm, n.s.) but geometric-valid degrades strongly ($-12.5/-14.0$ pp at $2/5$ mm, -41.5 pp at 10 mm)
Interface clearance & tolerance	Physical insertability of the fastener		insert	sensitive continuous feature; -54.5 pp reward at 0.75 mm clearance
Construction state & sequence	Gate whether the action is legal now		precondition check	discretely fatal (0%)
Physical properties (mass, friction)	Force-level and stability	contact	contact stages	<i>scaffolded</i> : no dynamical channel in the transport-and-insert executor (bit-identical outcomes); not ranked

Table S2: Provenance of task information: source class, extraction, runtime use, and repair responsibility. IFC-native = present in the delivery; derived = computed from IFC geometry/material; task annotation = CSR property-set/robot-task specification; assumed = default that must be escalated, never silently trusted.

Field		Source class	Extraction	Runtime use	Repair source
Component identity / IFC GUID		IFC-native	preserved through compilation	cross-reference closure	IFC source
Member pose (absolute)	pose	IFC-native	<code>IfcLocalPlacement</code> → USD transform	transport target pose	IFC source
Interface ordinates (hole, member-relative)	co-ordinates	derived	IFC hole feature → evidence record	fine alignment at hole mouth	evidence record
Interface clearance / tolerance	clearance	task annotation	CSR property set → evidence record + provenance	geometric insertability gate	evidence record
Construction state / sequence	state	task annotation	task state machine	precondition gate	task file
Mass / inertia	inertia	derived	IFC volume × material density (+uncertainty)	physics stability	derived value
Work zone / payload	zone	task annotation	CSR property set	multi-scale activation	task file
Robot capability / preconditions	capabilities	task annotation	robot-task specification	executor configuration	task file
Unannotated required field	required field	<i>assumed</i>	none (absent)	none → escalate	human review

Table S3: Construction-scene readiness contract (v0.3.0). The frozen v0.2.0 set carries 30 rules (CSR-SCN-001–030); v0.3.0 adds CSR-SCN-031 and CSR-SCN-032 for evidence-grounded interface fidelity, 32 rules in total. The versioned registry is exported at `docs/scene-readiness-rules.csv`, and per-scenario coverage is reported in the supplementary material.

Contract category	Required scene evidence	Deterministic check	Severity	Consequence if violated
Identity and closure	IFC GUID, USD prim path, closed task and asset references	Uniqueness and reference resolution	Critical	Wrong or unresolved task object
Spatial consistency	Units, up axis, frames, transform chain, work-zone membership	Transform normalization and bounds	Critical	Unreachable or misplaced interaction
Assembly interface	Paired interface, insertion axis, tolerance, grasp and support regions	Geometric and relational constraints	Critical	False alignment or infeasible insertion
Construction state	Current state, legal predecessor, action precondition, completion criterion	State-machine validation	Critical	Unsafe or meaningless action
Physics and stability	Mass, inertia, collision, contact property, temporary support	Physical admissibility and support tests	Critical or major	Unstable or non-physical simulation
Provenance and fidelity	Property source, confidence, uncertainty, activated payload and level of detail	Authority and task-fidelity policy	Major	Unauditable result or hidden interaction error

Table S4: Benchmark composition. Faulted-case counts are read from the frozen partition manifest (`data/partition-manifest.json`): 9 development, 24 test, and 2 challenge faults per scenario scene, with test and challenge faults evaluated on held-out geometric variants. All scenarios are exercised end-to-end at the compile–validate–repair level; robot execution in this paper is reported for the pin-insertion scenario at the fixed-base level, and the remaining scenario $\times$ level combinations are the benchmark’s intended progression.

Scenario	Parameterization	Robot task levels	Faulted cases	Primary outcome
Connection-plate positioning	Component size, pose, target height, work-zone obstacle	Fixed whole mobile; base; body;	9 dev / 24 test / 2 challenge	Placement pose and state completion
Simplified pin insertion	Interface axis, clearance, friction, initial misalignment	Fixed whole mobile; base (executed here); whole mobile (progression)	9 dev / 24 test / 2 challenge	Insertion depth without jam
Sequential assembly	Component order, temporary support, path obstruction, state transition	Fixed whole mobile; base; body;	9 dev / 24 test / 2 challenge	Complete legal task sequence

Table S5: Ablation and sensitivity analysis relative to the fixed-rule row of the main-text detection–repair table (105 paired cases each; bootstrap 95% CIs in brackets). The first four columns are measured on the ablated pipeline itself. The last column is *not* an ablation effect: it is the index of the *related* defect-consequence condition from Section 7.3 of the main text that conceptually matches the removed information, shown only to indicate how consequential that information family is at runtime (removing provenance is not the same as corrupting a physical value, and removing the task validator is not the same as every scene having a wrong task reference).

Ablation	Removed information or control	Critical recall	Valid repair	Resource use	Related defect outcome
No provenance	Source authority and uncertainty	0.894 [0.839, 0.941]	85.7% [79.0, 92.4]	unchanged	87.0% (physics, non-consequential)
No task validator	Task binding and action preconditions	0.787 [0.706, 0.868]	80.0% [72.4, 87.6]	unchanged	0.0% (task reference, fatal)
All layers loaded	Task-triggered payload activation	1.000	97.1%	+27% prims, +20% layer bytes	n/a
Fixed repair order	Agentic tool selection	1.000	97.1%	unchanged	identical to fixed rule
Free-form diagnostics	Typed diagnostic schema	1.000 (given)	89.0% [86.3, 91.6]	unchanged	n/a

Table S6: Repair outcome by frozen partition (development 27 / test 72 / challenge 6 cases; agent rows 5 seeds per case). Development cases are repaired to pristine equivalence by both grounded methods and test cases nearly all (fixed-rule 100.0%, constrained agent 99.4%); the challenge cases—whose repair would exceed the available engineering evidence—are split three repaired and three escalated by design, and never incorrectly repaired. The unconstrained agent fails across all partitions with a substantial incorrect-repair share.

Method	Dev valid (27)	Test valid (72)	Challenge valid (6)	Challenge escalated
Fixed-rule conver- sion	100.0%	100.0%	50.0%	50.0%
Validator-grounded agent	100.0%	99.4%	50.0%	50.0%
Unconstrained agent	40.0%	32.8%	0.0%	80.0%

Incorrect repair is 0.0% in every partition for the fixed-rule and validator-grounded rows; the unconstrained agent’s incorrect share is 19.3%/24.2%/20.0% (dev/test/challenge).

Table S7: Runtime resource use and task-level outcomes. Resource cells are reported per composition mode on the held-out variant scenes (structural metrics are exact; load time is the median of five cold opens and varies with host). Peak memory stayed below the 1 MB measurement threshold. Task stages are measured with the frozen RL executor on the defect-free scene (the cell reports the seed-42 partition; the seed-123 value is 84.0%, Table S8); planning, grasp, and alignment are not instrumented separately under the virtual-grasp scaffold (Section 7.3 of the main text).

Condition	Active prims	Peak memory	Load time	Planning	Grasp	Alignment	Insertion	Full task
Direct/all loaded	28–30	<1 MB	1.4 ms	n/a	n/a	n/a	n/a	n/a
Fixed rule	identical compiler output as full method				n/a	n/a	n/a	n/a
Free-form agent	method-independent compilation output; task cells not instrumented (Section 7.3 of the main text)							
Full/task activated	22–24	<1 MB	1.5 ms	n/a	n/a	n/a	87.0% (seed 42)	= insertion

Table S8: Per-condition geometric-valid statistics behind the main-text confusion matrix: 200 paired episodes per condition and seed partition, defect-free versus defect scene geometric-valid successes (ok→def), discordant pairs b/c (lost/gained), and exact two-sided McNemar p ; (a) seed 42, (b) seed 123. *Validator* names the rule that fires on the condition’s contract-equivalent fault (v0.2.0; mapping in the text); *silent* conditions violate no declared rule. Cells: TP = flagged and consequential; FN = silent but consequential; TN = silent, no significant degradation detected in the batch; FP = flagged without detected degradation (none occurred). Generated from the per-episode ledgers of the main-text defect-consequence table.

(a) seed 42					
Condition	rule	ok→def	b/c	p	cell
hole +2 mm	silent	87→62	44/19	0.0022	FN
hole +5 mm	silent	87→59	53/25	0.0020	FN
hole +10 mm	silent	87→4	85/2	4.9e-23	FN
member +10 mm	silent	87→83	29/25	0.68	TN
clear. 0.75 mm	silent	87→65	41/19	0.0062	FN
clear. 0.5 mm	silent	87→41	52/6	3.2e-10	FN
clear. 0.25 mm	silent	87→15	74/2	7.7e-20	FN
sem. missing	SCN-003	87→2	85/0	5.2e-26	TP
sem. wrong	SCN-004	87→0	87/0	1.3e-26	TP
task ref.	SCN-028	87→0	87/0	1.3e-26	TP
state	SCN-021	87→0	87/0	1.3e-26	TP
mass ×0.5	silent	87→87	0/0	1.0	TN
mass ×2	silent	87→87	0/0	1.0	TN

(b) seed 123					
Condition	rule	ok→def	b/c	p	cell
hole +2 mm	silent	86→77	39/30	0.34	TN
hole +5 mm	silent	86→62	53/29	0.011	FN
hole +10 mm	silent	86→11	82/7	2.4e-17	FN
member +10 mm	silent	86→74	41/29	0.19	TN
clear. 0.75 mm	silent	86→57	42/13	1.1e-4	FN
clear. 0.5 mm	silent	86→36	58/8	1.8e-10	FN
clear. 0.25 mm	silent	86→11	76/1	1.0e-21	FN
sem. missing	SCN-003	86→0	86/0	2.6e-26	TP
sem. wrong	SCN-004	86→0	86/0	2.6e-26	TP
task ref.	SCN-028	86→0	86/0	2.6e-26	TP
state	SCN-021	86→0	86/0	2.6e-26	TP
mass ×0.5	silent	86→86	0/0	1.0	TN
mass ×2	silent	86→86	0/0	1.0	TN

(a) faulted evidence record (friction)

```
{
  "entity": "default_steel_contact",
  "property": "friction",
  "source": {
    "benchmark_protocol": "steel_contact_v0.1",
    "source_type": "unverified_model_generation",
    "value": 0.5,
    "unit": "dimensionless",
    "confidence": 0.2,
    "uncertainty_interval": [
      0.3,
      0.7
    ],
    "authority_level": 1,
    "auto_repair_allowed": false,
    "requires_human_confirmation_for_deployment": true
  },
  "t": true
}
```

(b) typed validator finding (CSR-SCN-023)

```
{
  "rule_id": "CSR-SCN-023",
  "severity": "critical",
  "path": "$.provenance[7].source_type",
  "message": "A critical physical property cannot rely solely on an unverified generated value."
}
```

(c) validator-grounded agent escalation

```
{
  "method": "validator-grounded-agent",
  "case": "connection_plate_positioning_v001_unsupported_generated_friction",
  "outcome": "escalated",
  "episodes_escalated": "15/15 (3 scenes x 5 seeds)",
  "note": "no authoritative friction source; repair refused rather than invented"
}
```

Figure S1: Auditable failure–diagnosis–escalation sequence for the held-out challenge case `unsupported_generated_friction` (`connection_plate_positioning_v001`). (a) The faulted provenance record: friction entered by unverified model generation with low confidence and repair blocked by policy. (b) The deterministic typed finding (CSR-SCN-023). (c) The frozen validator-grounded model escalates in all 15 episodes rather than inventing a friction value. All panels render committed artefacts (scene JSON, validation report, episode ledger) via `scripts/make_case_study_figure.py`.